Neural-Symbolic NLP on a Small Transformer: Towards Neurosymbolic World Models for Manipulation

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***Abstract*—Neural networks trained by next-word prediction are strong at perception, but keep no explicit link to the rules behind a decision. Sheth, Roy and Gaur argue that safety-critical AI needs knowledge set as guardrails during training, not after it, and sort neurosymbolic methods into *lowering*, which compresses symbolic knowledge into the network, and *lifting*, which maps neural outputs to symbols for reasoning. On the lowering side, the semantic loss of Xu et al. turns a logical constraint into a training term. Ahmed et al. extend it to autoregressive models, where the exact probability of a constraint is #P-hard, and on GPT-2 pair it with word banning at decoding. A loss gives no test-time guarantee, and a mask leaves the model unchanged. This report asks what happens when one knowledge base drives both and a verifier lifts the output back to symbols, on a small transformer, and proposes the same split for neurosymbolic world models in robotic manipulation.**

***Index Terms*—Neuro-symbolic AI, lowering and lifting, semantic loss, constrained decoding, knowledge bases, world models, robotic manipulation.**

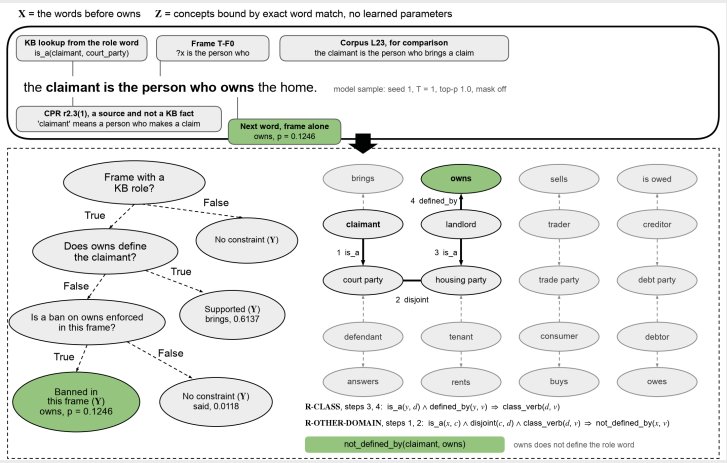


Fig. 1. **The core logic.** One sampled sentence, "the claimant is the person who owns the home", traced through the civil-law KB: the frame matches, the KB holds a ban on *owns* through *is_a*, *disjoint*, *is_a* and *defined_by*, and *owns* is banned in this frame. Network alone: p=0.1246 on *owns*, 0.6137 on *brings*. One unmasked sample (seed 1, T=1, top-p 1.0); educational KB, not legal advice.

I. Introduction

Foundation models give robots broad priors but no guarantee that a step obeys a rule. Vision-language-action (VLA) policies above 90% success on standard LIBERO fall to 0.0% under simple perturbations [1]. On Towers of Hanoi, a PDDL-based neuro-symbolic system reached 95% against 34% for the best fine-tuned VLA [2].

The same gap appears in language at a scale small enough to inspect. A language model trained on next-token cross-entropy has no notion of truth, because the loss only rewards text that looks like the corpus. Sheth et al. [3] frame the remedy as an integration of neural perception with symbolic knowledge, in two families: *lowering* compresses knowledge into the network, and *lifting* extracts neural outputs into symbols for reasoning. For mental-health assistance, they report 70% expert satisfaction for an intertwined pipeline against 47% for an LLM in a federated one; the process-knowledge line [4], [5] argues that such systems must follow, and explain themselves by, the expert's own guideline. On the lowering side, the semantic loss [6] measures how far the network's output is from satisfying a logical constraint, and the pseudo-semantic loss [7] extends it to autoregressive models.

This project is a from-scratch TypeScript pipeline (no ML libraries) that climbs from a rule-based chatbot to a GPT-style transformer, plus a symbolic layer on top: a knowledge base (KB), a hard logit mask at decoding and a semantic loss in training. `w3cj/how-llms-work` [8] is the reference architecture: same five stages, and no code copied. It specifies one pipeline in which the same KB shapes training through a semantic loss and constrains decoding through a hard logit mask. It measures the baselines that motivate the pipeline. One full-length run, at one seed and λ_s=50, has been compared with the baseline; of the planned experiments, E1, E2 and E3 have been run at four training seeds, and E4 without its held-out measures. Every number measured in this project comes from code, not typed in by hand; figures quoted from papers are attributed to them. The status of each stage is in Fig. 2 and Table I.

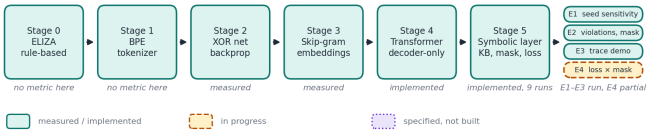


Fig. 2. Pipeline and status of each stage. Teal: built or run; amber: in progress (E4, without held-out measures).

Try it. The demo takes one parameter line, for example `300 0.8 0.9 2 40 2 42`: epochs, temperature, top-p, layers, max tokens, λ_s and seed. It trains on the server and streams every epoch. Send `2000 0.8 0.9 2 40 50 42` to replay the full-length run from its saved weights. The free host sleeps when idle, so the first request can take a minute or more.

Research report for EPSRC DLA project S3.5-COM-Valentino, University of Sheffield, for Dr Marco Valentino (NLP group) and Dr Amir Ghalamzan. Full-length version; the 3-page version is `proposal.pdf`. Live demo: https://neural-symbolic-nlp.onrender.com/train-neural-symbolic. Code: https://github.com/nhan03bit/neural_symbolic_nlp.

TABLE I
STAGES OF THE PIPELINE.

Stage	What	State
0	ELIZA pattern-matching chat [9]	Works
1	BPE tokenizer [10]	Works
2	XOR network, 2→4→1 with backprop [11]	Works, measured
3	Skip-gram word embeddings [12], [13]	Fixed (was collapsed), measured
4	Decoder-only transformer [14]	Done (forward, backward, Adam, top-p generation, multi-threaded training)
5	Symbolic layer	Implemented: KB, hard mask, verifier, semantic loss trainer, lifting. One full-length run, one seed
6	Experiments E1–E4	E1, E2, E3 run at four training seeds; E4 run except its held-out measures

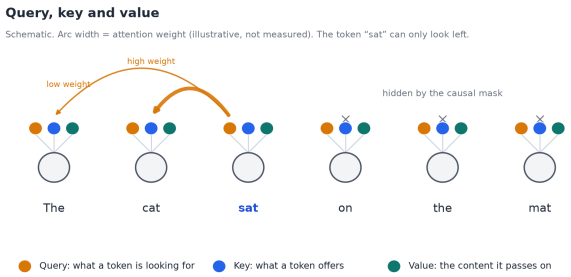


Fig. 3. Query, key and value (schematic).

II. METHOD

A. The transformer

Each token gets a query, key and value; attention weights are $\mathrm{softmax}(QK^\top/\sqrt{d_k}+M)V$ with a causal mask M; several heads run in parallel; a feed-forward net refines each token; blocks stack (Fig. 3). The block is pre-LayerNorm [15], as built in GPT-2 (Fig. 4):

$$x' = x + \mathrm{MHA}(\mathrm{LN}_1(x)),$$
$$x'' = x' + W_2\,\mathrm{ReLU}(W_1\,\mathrm{LN}_2(x') + b_1) + b_2.$$

Config: context 32 (training windows of 16 tokens), d=32, 2 heads, feed-forward width 128, 1–6 layers (default 2), Adam [16] lr 0.001, one full-batch step per epoch. That is **12,704 parameters per block**. With the whole-word BPE vocabulary built from the civil-law text (937 tokens, 1781 merges) the model has 87,401 parameters at 2 layers and 138,217 at 6. Training signal is next-token cross-entropy, nothing else. The implementation is in Table II.

Verification gates (5 pass): finite-difference gradient check (error $< 10^{-3}$), initial loss $\approx \ln V$, overfit a single sequence, causal-mask test, same seed gives same weights. **Bugs found and fixed.** The gates needed three forward/backward fixes in `transformer.ts`, made on 2026-11-02: the first residual multiplied instead of adding, attention skipped the token itself ($j < i$), and the gradient for K used K instead of Q. Older

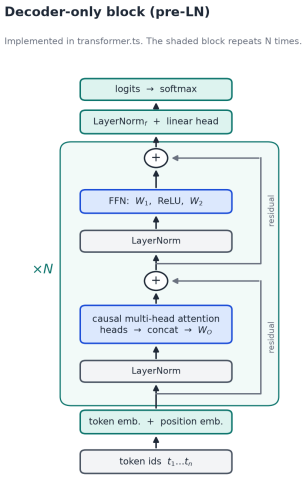


Fig. 4. Transformer block (pre-LN, as in GPT-2).

TABLE II
IMPLEMENTATION, IN `SRC/ROUTES/TRAIN-TRANSFORMER/`.

File	Role
`matrix.ts`	`matmul`, `matmulTransA/B`, Xavier init, seeded `rand` on flat row-major `Float32Arrays`
`transformer.ts`	LayerNorm, causal multi-head attention, FFN, block forward/backward, cross-entropy, Adam, top-p generation
`weigth-layout.ts`	Packs all weights into one `SharedArrayBuffer`; zero-copy views for every thread
`train-worker.ts`	Worker thread: forward + backward on its slice of sequences, gradients into shared memory
`train.ts`	BPE vocab, sliding windows, spawns `cpus−2` workers, sums gradients, Adam step, samples, checkpoints every 100 epochs
`index.ts`	`POST /train-transformer`, streams `init`, `epoch` and `done` events over SSE

weights stay on disk but are not loaded, because filenames now carry the model version.

B. One knowledge base, two integration points

The two designs are not alternatives. A KB of triples plus Horn rules gives one allowed set A_t per position, and that set is used twice (Fig. 5). Training shapes the distribution, so the model should learn to put little mass on banned tokens. Decoding guarantees the covered cases at test time. The published name for this pattern is constrained learning plus constrained inference. The closest published precedent is Ahmed, Chang and Van den Broeck [7], on GPT-2 (124M parameters) and RealToxicityPrompts: a pseudo-semantic loss with domain-adaptive training, then Word Banning at decoding. The combination has the lowest toxicity: 9.20% over all prompts, against 9.80% without banning and 14.05% for domain-adaptive training alone. Two differences matter: they score toxicity with the Perspective API rather than the constraint itself, and their loss works around a model sample, whereas the semantic loss here is a per-position term on corpus prefixes.

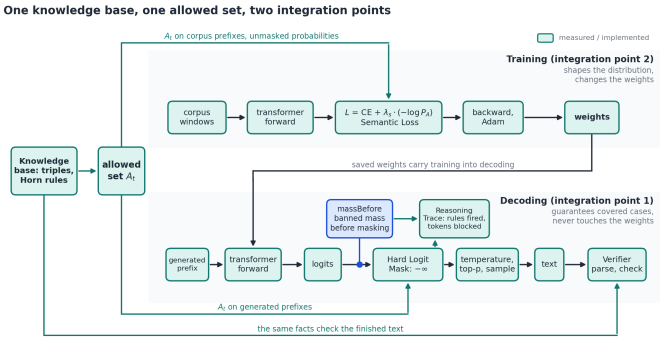


Fig. 5. One KB gives one allowed set, used by the semantic loss in training and by the hard mask at decoding; *massBefore* marks where the trained model meets the mask.

The KB covers definitions in the civil-law corpus, for example

$$\begin{aligned} &\mathit{counterpart}(\text{claimant}, \text{defendant}) \\ &\quad \wedge \mathit{defined_by}(\text{defendant}, \text{answers}) \\ &\qquad \Rightarrow \mathit{not_defined_by}(\text{claimant}, \text{answers}). \end{aligned}$$

Forward chaining closes the KB once, and a trigger turns a derived fact into banned token IDs when the context ends in a frame such as "the claimant is the person who". Every ban can be printed as a proof. Fig. 1 traces one unmasked sample from the saved model (seed 1, T=1, top-p 1.0), which began "the claimant is the person who owns the home", where the corpus says "brings a claim". The frame *?x is the person who* binds *claimant*, and the closed KB holds *not_defined_by*(claimant, owns) by R-OTHER-DOMAIN, so *owns* is banned. Class labels such as *court_party* are the KB's own, pairwise disjoint as labels and not as people; *defined_by*(landlord, owns) is a corpus simplification. These are read-only probes, not a measured effect: with the frame alone and the mask off, *owns* has p=0.1246, and 265 of 2,000 sampling seeds drew it at top-p 1.0; at top-p 0.6 all 2,000 drew *brings*. A test on the real corpus and tokenizer shows the triggers fire at 9 of 5,390 positions and never ban the next token in the corpus. The layout follows Sheth et al. [3], Fig. 4, and the drawing is original.

Method 1: the KB enforced at decoding (hard logit mask). At each step, the logits of banned token IDs are set to $-\infty$ before temperature and top-p. Optional companions are the M2 soft bias (towards supported tokens), the M3 reasoning trace (rules fired, tokens blocked and *massBefore*) and the M4 verifier (parses finished text into triples and checks them).

- *Guarantee.* Zero mass on covered banned tokens, but only relative to KB coverage, only for whole-word token IDs, and only when a trigger fires.
- *Limits.* It does not change the weights or choose among allowed tokens. A wrong rule is enforced absolutely. Errors concentrate in the translation from text to logic, so the verifier reports only on what it could parse.
- *Taxonomy.* In [3] this is 1a-style lowering, by analogy, not strict 1a: the knowledge is neither compressed nor learned. The mapping is ours.

Method 2: semantic loss in training. The objective is $L = \mathrm{CE} + \lambda_s(-\log P_A)$, where P_A is the probability mass on allowed tokens ([6]; the categorical form is [7], Eq. 3). It is a deterministic function of the logits; the randomness comes from seeded initialisation and top-p sampling, and this term leaves both in place.

- *It cannot see the choice among allowed tokens.* $H(p) = H_b(P_A) + P_A H(q_A) + (1-P_A)H(q_B)$, where q_A and q_B are p renormalised over allowed and banned tokens. The loss depends only on P_A, so $H(q_A)$ is outside its objective (our derivation). A constraint "only specifies what worlds are possible but does not specify which one is correct" [17].
- *It is a soft penalty.* Where data and rule conflict, the optimum at one context keeps banned mass $f_B/(1+\lambda_s)$, where f_B is the banned tokens' frequency in the data (our derivation). There is no test-time guarantee [18].
- *It acts only where it is applied.* Where the target is allowed, CE = masked CE + $(-\log P_A)$, so the term changes the speed of learning there, not the optimum (our derivation). It fires at 86 of 85,984 training-window positions (0.100%), with the target allowed at all of them, so a null result is plausible.
- *It may amplify seed differences.* Zombori et al. [19], Theorem 4, prove for softmax regression that, with no other supervision, this form of loss concentrates mass on whichever allowed token starts largest. Whether that happens in a transformer has to be measured.
- *Taxonomy.* Towards 2b (intertwined), not end-to-end differentiable reasoning: the KB is discrete and only the loss term is differentiable. Sheth et al. do not use the term "semantic loss", so this placement is also our reading.

Why the semantic loss needs the mask.

- *Violation rate cannot show what the loss adds.* With the mask on, violations of token-expressible rules should be 0 by construction; E4 found one surface form the mask does not cover ("owed", Sec. III-E). So the evaluation reads its contribution from *massBefore*, violations with the mask off, violations the verifier finds that a token mask cannot express, and held-out cross-entropy on targets that do not conflict with the KB.
- *Prediction, not a result.* A model trained with the loss puts less mass on banned tokens (P5), so the mask has to intervene less. The distortion at one context is $KL(q_A \| p) = -\log P_A$, the value of the semantic loss there (our derivation). A covered context still samples from q_A, so no finite λ_s makes the mask unnecessary for a correct rule.
- *Shared failure mode.* A wrong or missing rule enters both stages at once. If lifting were ever allowed to add bans, a wrong accepted triple would reach both stages, so that step stays off.
- *Perplexity cost.* A hard mask gives infinite perplexity on reference text containing a banned token [7]. Here,

TABLE III
MECHANISMS AND THEIR POSITION IN THE TAXONOMY.

Mechanism	What it does	Position
M1 Hard logit mask	Sets logits of KB-violating tokens to $-\infty$ before temperature and softmax	1a-style lowering, by analogy
M2 Soft bias	Adds $+\lambda$ to tokens the KB supports	1a-style lowering, by analogy
M3 Reasoning trace	Per step: rules fired, tokens blocked, mass before and after	Part of 2a
M4 Verifier	Parses generated text into triples, checks them against the KB	2a decoupled lifting
M5 Semantic loss	Training term rewarding mass on allowed tokens	Toward 2b

TABLE IV
LIFTING: STATUS OF A CANDIDATE TRIPLE.

Status	Meaning	Action
confirmed	KB already holds it	None. Evidence the network learned the fact, though on this model it may be memorisation
conflict	KB derives the opposite (*not_defined_by*)	Report with probability and proof. Direct evidence of the reasoning gap
novel	Neither	Accept only if $p \geq \tau$ **and** it creates no new contradiction

E2 computes perplexity from the weights alone, and E4 reports held-out cross-entropy on targets that do not conflict with the KB.

The stated target stays **zero mass on rule-excluded tokens and a calibrated spread over the rest**, with each guarantee stated next to its coverage.

C. Mechanisms and the taxonomy

In the terms of Sheth et al. [3], **lowering** compresses symbolic knowledge into the neural side, **lifting** lifts neural output into symbols (Fig. 6, Table III, Fig. 7). A mask is not understanding: it blocks a class of outputs and leaves the weights unchanged. The trace explains the symbolic decision, not the network's internal computation.

D. Lifting: network patterns into symbolic knowledge

This is the "neural patterns lifted into symbolic knowledge" half of the design. It probes the trained network with templates, reads its next-word distribution and turns confident verb predictions into **candidate** triples (Table IV). **Reported only.** The neural-symbolic trainer runs lifting at every epoch and shows each candidate's status, but a *novel* word that passes the gate is only flagged, never added to the KB. An accepted triple is the network's belief, not ground truth, and it would stay out of the mask and the semantic loss.

E. Two separate trainers

The baseline and the neural-symbolic model are trained apart, so the difference between them can be inspected directly (Table V). They share the corpus, tokeniser, model code, Adam and hyperparameters; only the loss differs. The compare view runs both models on one prompt and sampling seed, with the mask off and on, and shows their text, the mask trace, the verifier's claim counts and the banned mass at each definition frame. A test shows that at λ_s=0 and seed 42 the new trainer saves the baseline's weights and losses, so λ_s=0 is the CE-only control at seeds the baseline does not offer.

TABLE V
THE TWO TRAINERS.

	Baseline	Neural-symbolic
Code	`train-transformer/train.ts`	`neural-symbolic/train.ts`
Page	`/train-transformer`	`/train-neural-symbolic`
Loss	CE	CE $+\lambda_s(-\log P_A)$
Seed	42 only	Chosen per run
Weights	`transformer-weights-`$\star$	`neural-symbolic-weights-`$\star$, named with λ_s, seed and KB hash

TABLE VI
FALSIFIABLE PREDICTIONS.

#	Prediction	Test	Refuted if
P1	With the hard mask on, the violation rate is 0 for token-expressible constraints	E2	any such violation
P2	With the processor off, output equals the unconstrained baseline at the same seed	unit test	any difference
P3	The unconstrained model violates the KB at a non-trivial rate	E2	rate near 0
P4	Seeds give similar loss but different weights	E1	seeds agree closely in weights
P5	Semantic loss lowers the network's own forbidden mass	E4	not lower, mask off
P6	Semantic loss reduces solution ambiguity across seeds but not weight ambiguity	E4	JS of $(P_A, 1-P_A)$ does not shrink, or weight cosine near 1
P7	The cost on unconstrained text is small	E4	large rise in perplexity where no rule fires

F. Falsifiable predictions and planned experiments

P1 and P2 follow from the construction, so they test the implementation; P3–P7 are empirical and may fail (Table VI). Table VII lists what each experiment has and still needs.

III. RESULTS

A. Baselines before the symbolic layer

Everything in this subsection was trained by next-token or context prediction alone, with **no** KB, mask, bias, trace, verifier or semantic loss.

XOR: same function, unrelated weights. A single layer cannot learn XOR (loss 0.2845 $\rightarrow$ 0.2842). A 2$\rightarrow$4$\rightarrow$1 network can (Table VIII, Fig. 8). At 5000 epochs all seeds compute the same function, but the first hidden unit's weights differ completely, for example $[5.84, -1.49, 6.18, -0.28]$ for seed 42 against $[0, -6.50, -1.93, 5.29]$ for seed 1.

Skip-gram: the civil-law vocabulary, and a wrong answer. The first version gave every word pair cosine 1.0. Five independent bugs were the cause: an off-by-one in the sampling table, a missing positive update, the wrong row

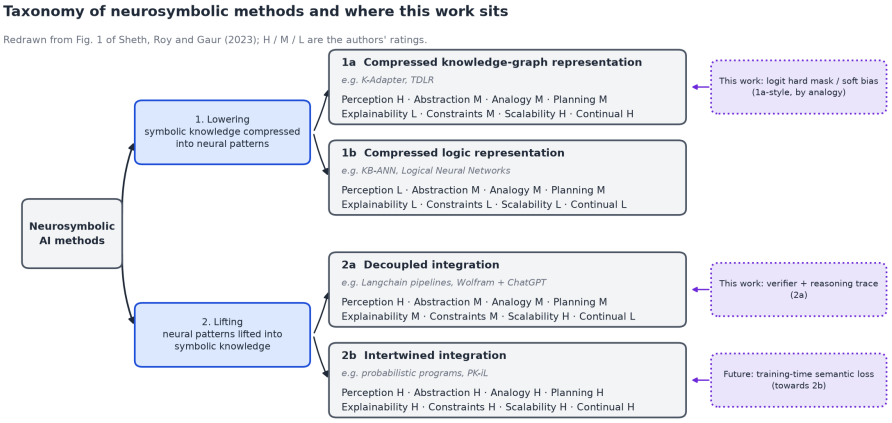


Fig. 6. Taxonomy mapping, redrawn from [3], Fig. 1 (H/M/L: its authors' ratings). *Status note:* the "Future" semantic loss is now implemented.

TABLE VII
EXPERIMENTS: WHAT IS IN PLACE AND WHAT IS LEFT.

Experiment	Tests	In place now	Left to do
E1: seed sensitivity	P4	Run: λ_s=0 at seeds 42, 1, 2 and 3, 2000 epochs	Nothing for P4. Held-out text, not just held-out prompts, needs the held-out split
E2: violation rate, mask off/on	P1, P3	Run at training seeds 42, 1, 2, 3 (the λ_s=0 models), 24 prompts × 50 sampling seeds, mask off and on	Perplexity from the weights alone on held-out text. Close the "owed" gap in the mask or the verifier, then re-run
E3: one full trace	—	Run: one generation's trace for both models	Nothing
E4: loss × mask ablation	P5, P6, P7	Run: $\lambda_s \in \{0, 2\}$ × mask off/on × seeds 42, 1, 2, 3, 2000 epochs, probe contexts outside training and a frequency-matched control set	A held-out split in the neural-symbolic trainer, a held-out KB, and held-out cross-entropy (P7 here is measured on training text)

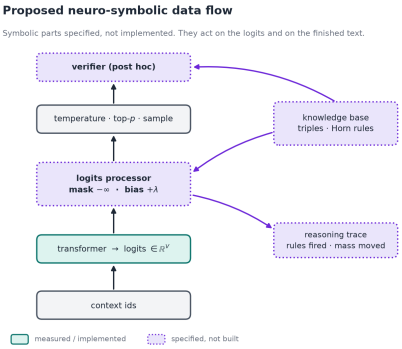


Fig. 7. Data flow at decoding. *Status note:* the parts marked "specified, not built" are now implemented.

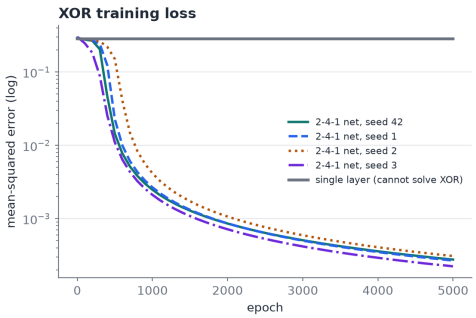


Fig. 8. XOR loss and weights across seeds.

TABLE VIII
XOR, FOUR SEEDS, SAME DATA AND SETTINGS (FINAL MSE).

Epochs	Seed 42	Seed 1	Seed 2	Seed 3
500	0.0144	0.0227	**0.1506** (not conv.)	0.0110
5000	0.00028	0.00027	0.00031	0.00022

updated for negatives, a wrong analogy sign, and a corpus join that never lower-cased. The model now trains only on England & Wales civil-dispute text (222 definitions plus 34 dispute stories, 401 sentences; vocab 924 whole-word tokens from 5000 BPE merges; dim 32, window 2, 17,142 training pairs; 10,000 epochs; seed 42). Words that share sentences end up close (Table IX, Fig. 9): the consumer-rights roles form one cluster, and the claimant sits next to the creditor because most claims in the corpus are debt claims. But none of the five analogies returns a role word (Table X). The model learned co-occurrence, not the relation "the party who supplies or lends, against the party who receives or owes".

Nearest neighbours. From the `/train-embed` page with `claimant defendant landlord tenant` (10,000 epochs, seed 42; loss at epoch 0 4.102625), Table XI and Fig. 10. Pairwise: landlord–tenant 0.59, defendant–tenant 0.54, claimant–defendant 0.44, defendant–landlord 0.44, claimant–tenant 0.43, claimant–landlord 0.40.

Four seeds: the geometry agrees, the coordinates do not.

TABLE IX
SKIP-GRAM COSINES, CIVIL-LAW CORPUS.

Pair	Cosine	Pair	Cosine
buyer / customer	0.76	defendant / tenant	0.54
seller / buyer	0.74	claimant / defendant	0.44
claimant / creditor	0.69	trader / consumer	0.42
landlord / tenant	0.59	trader / court	−0.02
creditor / debtor	0.58	judge / contract	−0.13

TABLE X
SKIP-GRAM ANALOGIES: NONE RETURNS A ROLE WORD.

Query	Expected	Returned (cosine)
landlord − tenant + consumer	trader	*renters* (0.57)
trader − consumer + tenant	landlord	*return* (0.65)
claimant − defendant + debtor	creditor	*whether* (0.61)
creditor − debtor + defendant	claimant	*between* (0.58)
seller − buyer + customer	supplier	*proves* (0.72)

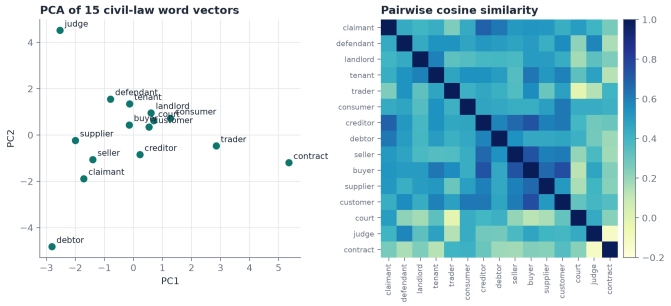


Fig. 9. PCA and cosine matrix of 15 civil-law word vectors, from the saved embeddings.

TABLE XI
FIVE NEAREST NEIGHBOURS (COSINE).

Word	Neighbours
claimant	creditor 0.69, remoteness 0.67, limitation 0.67, evidence 0.66, test 0.64
defendant	noise 0.69, loser 0.66, authorities 0.64, hearing 0.62, talking 0.62
landlord	tenant 0.59, protect 0.59, higher 0.57, thought 0.57, loss 0.56
tenant	income 0.67, area 0.66, disputed 0.64, buyer 0.63, price 0.62

Measured on the earlier toy corpus, not yet re-run on the civil-law corpus (Table XII, Fig. 11). The learned *function* is stable across seeds; the *weights* are not. So the internal solution is neither unique nor designed, and you cannot read a reason off it. Stability also does not mean correctness: the same analogy error appears on every seed.

Transformer baseline. 2000 epochs, 2 layers, temperature 1, top-p 0.6, 20 tokens, seed 42, sent as `2000 1 0.6 2 20` (about 86 minutes); Table XIII, Fig. 14, left. Sample at epoch 2000, prompt "the claimant is": *"the claimant is the person who brings a claim. the defendant is the person who answers a claim. the claimant asks"*. **This loss shows memorisation, not generalisation.** The whole sample is an unbroken stretch

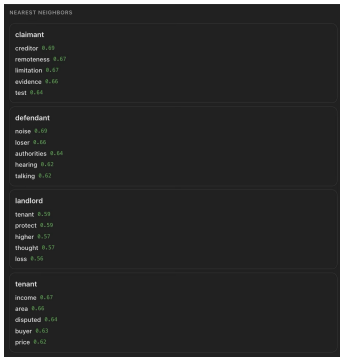


Fig. 10. Nearest neighbours, from the `/train-embed` page.

TABLE XII
SEED AGREEMENT, SKIP-GRAM, TOY CORPUS (6 SEED PAIRS).

Measure	Range
Cosine of the *same word* across seeds	−0.28 to 0.09
Correlation of the *pairwise geometry* across seeds	0.977 to 0.984
Analogies correct (our answer key)	5/7 to 6/7

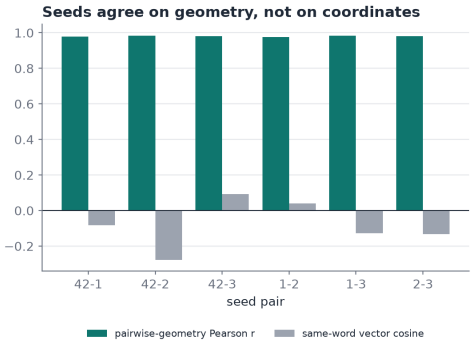


Fig. 11. Seed agreement: geometry agrees, coordinates do not.

TABLE XIII
TRANSFORMER BASELINE, 2000 EPOCHS, SEED 42.

Measure	Value
Training text	256 items (222 definitions + 34 stories)
Vocabulary	937 tokens (1781 BPE merges)
Parameters	87,401
Training windows	5,374 windows of 16 tokens
Loss at epoch 0	6.868 ($\ln 937 = 6.843$, a uniform guess)
Loss at 100, 300, 500, 1000, 1500	4.850, 2.217, 1.051, 0.391, 0.287
Lowest logged loss	0.248 at epoch 1980
Final loss, epoch 2000	0.2489 (perplexity ≈ 1.28)
Held-out loss	Not measured: trained on all 256 items

of the training text, and as the loss falls the samples turn into copies (Table XIV); a token counts as copied if it sits inside a run of 8 or more tokens found verbatim in the training token stream. A superseded 300-epoch baseline, recorded on 2026-10-31 with the three defects present, stalled at a loss of 5.678 (perplexity ≈ 292) with domain words in no order; it is kept only as a record.

What is measured so far: low loss is not correctness

TABLE XIV
GENERATED TOKENS COPIED FROM THE TRAINING TEXT.

Epochs	Samples	Copied, mean	Entirely copied
0–300	31	0.0%	0
310–500	20	5.5%	0
510–1000	50	79.7%	0
1010–1500	50	99.5%	15
1510–2000	50	100%	38

TABLE XV
MEASURED BASELINES (NO KB, MASK OR SEMANTIC LOSS).

Baseline	Measure	Value
Skip-gram, one seed	Role analogies correct	0 of 5
	landlord–tenant cosine	0.59
Transformer, 2000 epochs, all text	Final loss	0.2489
	Tokens copied, epochs 1510–2000	100%
Transformer, 4-fold held-out stories, 16 runs/config	Test loss, lr 3e-4 / 1e-4 (nats/token)	5.389±0.165 / 5.386±0.174
	Bigram / unigram baseline	5.290 / 6.012
	Runs beating the bigram, per character	0 of 16 / 1 of 16

(Table XV). Fluent samples from earlier in the run state definitions the corpus does not contain, and contradict its definition of a claimant as "the person who brings a claim": *"the claimant is the person who owns the home. the tenant is the person who rents the home. the landlord gets"* (epochs 1000 and 1370); *"the claimant is the person who rents the home. the landlord gets the rent. the tenant pays the rent. the"* (epoch 1380). These samples show that this happens, not how often. On held-out stories, a larger configuration (d=64, 4 heads) does not beat a bigram model on any of the four folds. The Skip-gram result is the same gap in another form: moderate similarity, wrong relation.

The reasoning gap. The objective has no notion of truth: it rewards sounding like the corpus. Weights come from a random start, and XOR and Skip-gram show equal behaviour from unrelated weights. "Pure emergent, not designed" means opaque: attention maps show where the model looked, not a justification it could be held to. More layers do not change the objective; whether reasoning emerges at scale is open, and we claim only that nothing in *this* small model asks for it or exposes it. The semantic loss has limits of its own, from the literature: Xu et al. [6] report that 69.89% of test predictions satisfied the constraint on grid paths and 55.28% on preference rankings, large gains but far from complete; a missing rule constrains nothing and a wrong one is trained in; and even with correct constraints, the generalisation benefit comes with a bias toward a suboptimal model [20]. The two sources of randomness also stay (Table XVI). Fixing the seed makes a run repeatable, not justified; greedy decoding removes the draw, but the output is then one fixed point of one seed's distribution. Removing all uncertainty would be the wrong goal anyway, because language has many acceptable continuations and ambiguity is a normal feature of it [21], [22].

B. One full-length run: λ_s=50 against the CE-only baseline

One seed, one λ_s, trained on all text. This describes two models; it does not measure an effect of the loss. λ_s=50 was chosen after 300-epoch trial runs at $\lambda_s \in \{0, 2, 10, 50\}$, simply as the highest value tried. Sent as `2000 0.8 0.9 2 40 50 42` (about 90 minutes). The control is the same trainer at λ_s=0; its epoch-2000 weights are identical to the baseline's, as are all 210 logged baseline losses. Table XVII is read from the saved weights at the 86 training positions where the KB fires (Fig. 12).

- **The banned mass is lower at every epoch above.** At epoch 2000, semantic loss is 73 times lower. The semantic loss logged during training peaks at 4.73e-4 (epoch 689) without the term and at 2.26e-5 (epoch 585) with it.
- **The early cost does not last in this run.** Up to about epoch 600 the term also slows the allowed words, including the role's own verb. By epoch 1000 it is ahead on both.
- **The training loss costs a little.** Final cross-entropy is 0.2524 against 0.2489; both models copy the training text, so this is not a held-out measure.

A rule counts as violated when its banned word's mean p at its trigger pattern is at least 0.02; there are 48 rules (Table XVIII). Across all 2001 logged epochs, CE only violates at least one rule at 1846 of them, with a maximum of 29 at epoch 529. The neural-symbolic model violates at least one at 175, with a maximum of 3 at epoch 585, and none after epoch 1746. CE only's 7 violations at epoch 2000 are all on the claimant and defendant frames, led by *claimant* → *answers* at p=0.124.

It acts where it fires, and not reliably beyond. Each role appears in only one of the three definition frames in the corpus, such as *trader* only in "is a business that". The compare view probes all 24 role-and-frame contexts with the prompt alone, mask off (Table XIX). Lifting shows the same split: at epoch 2000, conflicts on trained frames fall from 4 to 0; on untrained frames from 28 to 15, but nearly all confident ones ($p \geq 0.3$) remain, 9 against 10. These are mostly "sells" after "is a business that", at p 0.81–0.96 for 6 roles, which suggests the frame, not the role, drives the prediction; that reading is not tested separately. With the prompt "the claimant is the person who" (sampling seed 42, T=0.8, top-p 0.9), both models write the same text with the mask off and on; the verifier finds 2 supported claims and 0 contradicted, and the mask trace shows banned mass at step 0 of 0.3252 for the baseline and 0.0026 for the neural-symbolic model. Screenshots of the run are in Fig. 13.

C. Pre-symbolic transformer vs neural-symbolic run

Both runs have the same architecture, corpus and seed, and 2000 epochs (Fig. 14). The left run is the CE-only baseline,

TABLE XVI
THE TWO SOURCES OF RANDOMNESS, AND WHAT THE MECHANISMS (M1–M5) DO TO THEM.

Source	Where	What the approach does to it	What is left
Weight initialisation	Xavier init from the seeded `rand` (`matrix.ts`)	The semantic loss constrains outputs, not coordinates, so it adds a term to training that still starts from random weights. The mask, trace and verifier never touch the weights.	Seeds still end at unrelated weights, and may disagree more among allowed tokens [19]
Top-p sampling	The draw in `generateText`, from the same `rand`	The semantic loss changes the probabilities drawn from, and the mask removes banned tokens before temperature and top-p. The draw itself stays. The verifier only checks text after it is drawn.	The next token is still picked at random from what remains in the nucleus

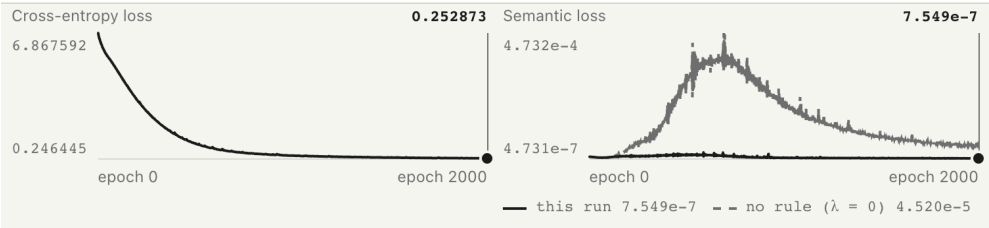


Fig. 12. Cross-entropy (left) and semantic loss (right) by epoch, λ_s=50; the λ_s=0 control is dashed and ends at 4.520e-5, against 7.549e-7.

TABLE XVII
λ_s=50 VS CE ONLY AT THE 86 CONSTRAINED POSITIONS.

Ep.	Model	Sem. loss	Banned	p(next word)	p(own verb)
300	CE only	7.73e-5	0.0725	**0.045**	**0.033**
	CE + SL	**9.85e-6**	**0.0098**	0.036	0.012
600	CE only	3.72e-4	0.2831	**0.339**	**0.249**
	CE + SL	**1.34e-5**	**0.0133**	0.236	0.140
1000	CE only	2.06e-4	0.1647	0.739	0.612
	CE + SL	**2.34e-6**	**0.0023**	**0.866**	**0.732**
2000	CE only	4.58e-5	0.0419	0.946	0.809
	CE + SL	**6.26e-7**	**0.0006**	**0.978**	**0.841**

TABLE XVIII
RULES VIOLATED (OF 48) AND LIFTED CONFIRMED / CONFLICT / NOVEL.

Epoch	Viol., CE	Viol., CE+SL	Lifted, CE	Lifted, CE+SL
100	0	0	0 / 0 / 1	0 / 0 / 3
200	5	0	1 / 6 / 57	1 / 1 / 56
300	5	0	2 / 8 / 91	1 / 2 / 90
400	23	0	7 / 32 / 134	3 / 9 / 115
600	26	1	7 / 43 / 54	7 / 18 / 79
1000	22	0	8 / 42 / 15	7 / 14 / 26
1500	11	0	8 / 35 / 5	8 / 14 / 8
2000	7	0	8 / 32 / 5	8 / 15 / 4

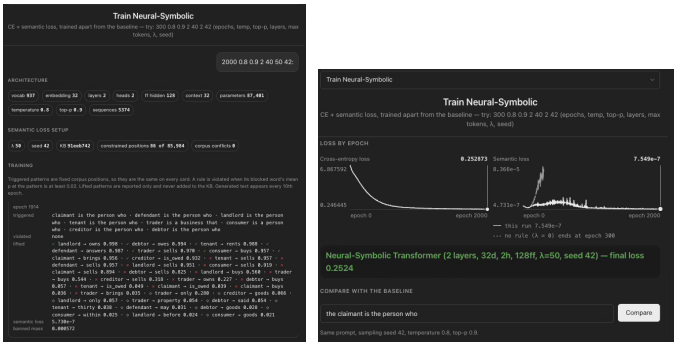
sent as `2000 1 0.6 2 20`, so its samples use temperature 1, top-p 0.6 and 20 tokens. The right run adds the KB and the semantic loss at λ_s=50, and samples at temperature 0.8, top-p 0.9 and 40 tokens. Final cross-entropy is 0.2489 on the left and 0.2524 on the right; on the right, no rule is violated at epoch 2000 (0 of 48).

D. E2 and E3 at one training seed

One training seed (42), so E2 is partial here; Sec. III-E completes it at four seeds. Everything below was fixed before running. *Models:* the CE-only baseline and the λ_s=50 model, both 2000 epochs. *Prompts:* every role in every definition frame, $8 \times 3 = 24$; 8 occur in the training text, and 16 put a role in a frame it never appears in. *Sampling:* seeds 0–49 per prompt, 20 tokens, at T=0.8, top-p 0.9 and at T=1, top-p 1. *Mask off:* the processor runs with λ=0 in bias mode, which edits nothing but records *massBefore*; all 4,800 mask-off texts matched plain generation (P2). *Violations:* claims the verifier parsed and found contradicted by the KB; the 95% Wilson interval treats claims as independent, so it is too narrow, and covers sampling only (Table XX).

TABLE XIX
PROMPT-ONLY PROBES OF ALL 24 ROLE-AND-FRAME CONTEXTS, MASK OFF.

Contexts	Baseline	Neural-symb.	Lower/higher
8 trained (role's corpus frame)	0.0959	0.0013	7 / 1
16 untrained (other two frames)	0.6154	0.4975	9 / 7
of which "is a business that", 7 roles	0.7787	0.8331	1 / 6



(a) Epoch 1914 (b) Loss to epoch 2000

Fig. 13. The λ_s=50 run: (a) no rule violated, lifted patterns marked; (b) final loss 0.2524, λ_s=0 control (dashed) to epoch 300 only.

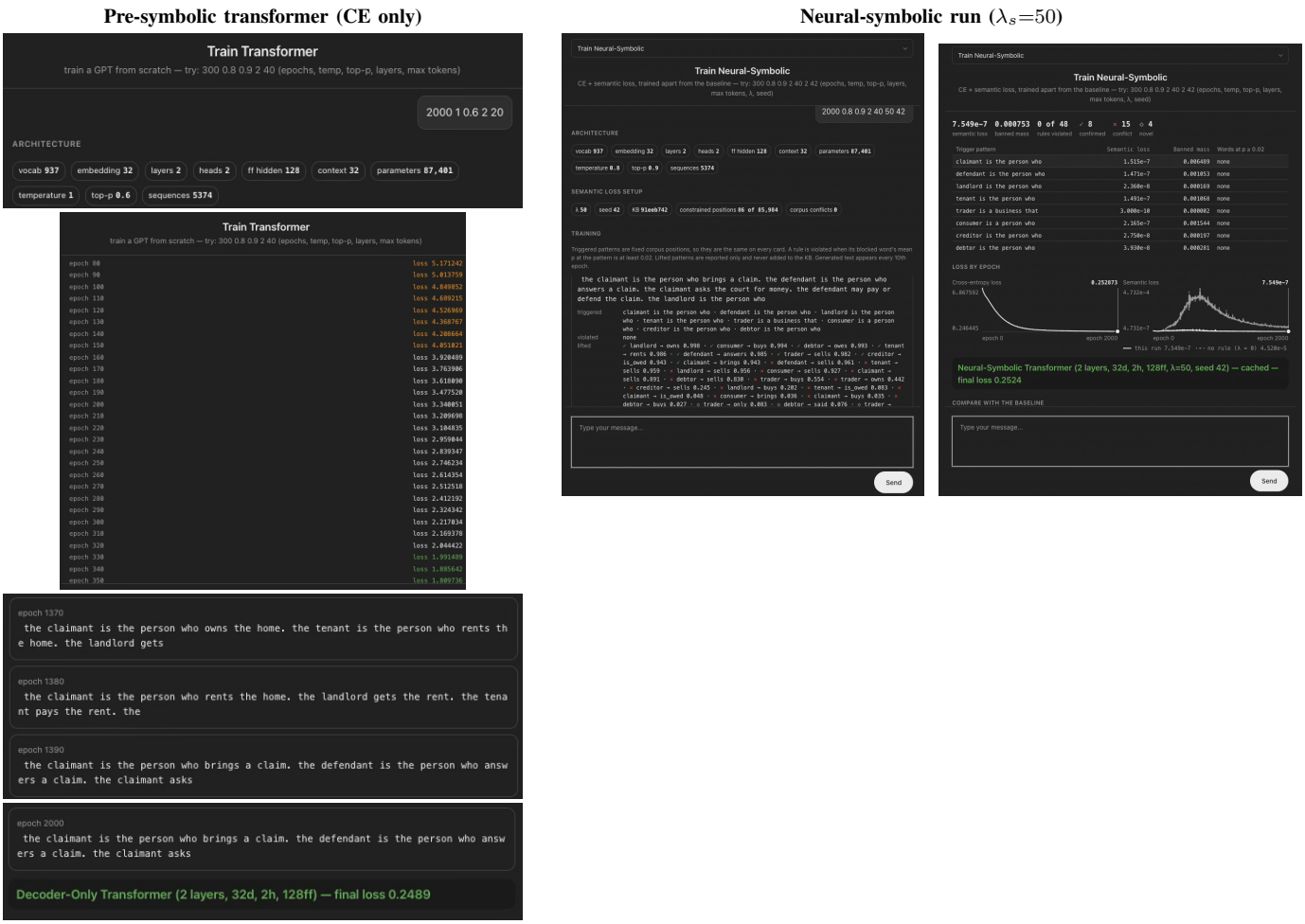


Fig. 14. **Pre-symbolic transformer vs neural-symbolic run.** Left, top to bottom: architecture, loss to epoch 350, samples at epochs 1370–1390, final loss 0.2489. Right: the live demo after 2000 epochs at λ_s=50, 0 of 48 rules violated.

TABLE XX
E2 AT SEED 42: CONTRADICTED / PARSED CLAIMS, WITH THE 95% INTERVAL. 400 SAMPLES PER CELL FOR THE TRAINING-TEXT PROMPTS AND 800 FOR THE OTHERS.

Prompts	Model	Mask off, T 0.8, top-p 0.9	Mask off, T 1, top-p 1	Mask on, both settings
8 in the training text	CE only	7 / 598 (1.2%, 0.6–2.4%)	28 / 578 (4.8%, 3.4–6.9%)	0 / 600, 0 / 578
8 in the training text	CE + SL (λ_s=50)	0 / 600 (0–0.6%)	0 / 573 (0–0.7%)	0 / 600, 0 / 573
16 not in the training text	CE only	515 / 1,253 (41.1%, 38.4–43.9%)	522 / 1,203 (43.4%, 40.6–46.2%)	0 / 938, 0 / 892
16 not in the training text	CE + SL (λ_s=50)	430 / 1,213 (35.4%, 32.8–38.2%)	420 / 1,159 (36.2%, 33.5–39.1%)	0 / 865, 0 / 839

- **With the mask on, no parsed claim contradicts the KB:** 0 of 5,885 claims across all eight cells. At this seed that checks the implementation (P1), not the model.
- **With the mask off, the CE-only model contradicts the KB:** 1.2% at T=0.8 and 4.8% at T=1 on the 8 training-text prompts, 41–43% on the 16 others.
- **On the training-text prompts, the λ_s=50 model made no contradicted claims in 1,173.** Its mean *massBefore* where a rule fired was 0.0010 and 0.0028, against 0.0739 and 0.0771 for CE only. These are training contexts, at one seed.
- **On the other 16 prompts, it still contradicts the KB in 35–36% of claims.** At T=0.8, 315 of its 430 contradicted claims are "sells" after "is a business that", the verb of trader, the one role trained in that frame; another 100 are "buys" after "is a person who", the verb of consumer. CE only does the same: 296 and 126 of its 515. Each model completes a frame with the verb of the role trained in it, whatever the role.
- **On those 16 prompts, the mask mostly turns contradicted claims into unknown ones.** For CE only at T=0.8, unknown claims rise from 50 to 443 and supported claims fall from 688 to 495.

E3: one full trace. Prompt "the claimant is the person who", sampling seed 42, T=0.8, top-p 0.9, mask on. Both models generated "the claimant is the person who brings a claim. the

TABLE XXI
E3: THE TWO STEPS WHERE A RULE FIRED.

Step	Context ends	Rules fired	Masked tokens	*massBefore*, CE	*massBefore*, λ_s=50	*massAfter*	Chosen
0	the claimant is the person who	6 bans (answers by R-COUNTER; owns, rents, sells, buys, owes by R-OTHER-DOMAIN), 1 support (brings)	answers, owns, rents, sells, buys, owes	0.3252	0.0026	0	brings
10	the defendant is the person who	6 bans (brings by R-COUNTER; owns, rents, sells, buys, owes by R-OTHER-DOMAIN), 1 support (answers)	brings, owns, rents, sells, buys, owes	0.0192	0.0002	0	answers

TABLE XXII
THE FOUR ARMS AT SEEDS 42, 1, 2, 3.

Arm (λ_s, mask)	Banned mass, 86 positions, mask off	Contradicted, 16 untrained prompts, T 0.8	Contradicted claims the mask did not cover	Held-out CE
CE only	0.024–0.049	35.4–47.9%	—	not measured (no held-out split)
CE + semantic loss	0.008–0.016	31.3–45.7%	—	not measured
CE + mask	— (mask edits after the network)	0% at 3 seeds, 17.8% at seed 1	167 at seed 1, all "owed"	not measured
CE + semantic loss + mask	—	0% at 3 seeds, 11.6% at seed 1	111 at seed 1, all "owed"	not measured

TABLE XXIII
E4 PER SEED (42, 1, 2, 3).

Measure	λ_s=0	λ_s=2
Banned mass, 86 constrained positions, mask off	0.042, 0.049, 0.024, 0.029	0.010, 0.016, 0.009, 0.008
Banned mass, 8 training-text prompts	0.096, 0.087, 0.036, 0.076	0.021, 0.024, 0.015, 0.014
Banned mass, 16 untrained prompts	0.615, 0.451, 0.734, 0.663	0.529, 0.432, 0.708, 0.532
Frequency-matched control set, untrained prompts	0.0009, 0.0052, 0.0003, 0.0039	0.0079, 0.0058, 0.0000, 0.0013
Contradicted claims, untrained prompts, mask off, T 0.8	41.1%, 35.4%, 47.9%, 46.0%	32.6%, 31.3%, 45.7%, 40.1%
$H(q_A)$ at constrained positions (bits)	0.127, 0.209, 0.167, 0.185	0.125, 0.199, 0.152, 0.159
Perplexity where no rule fires (training text)	1.283, 1.281, 1.268, 1.266	1.281, 1.281, 1.264, 1.266
Final training cross-entropy	0.2489, 0.2471, 0.2374, 0.2358	0.2478, 0.2471, 0.2340, 0.2356

TABLE XXIV
E1, λ_s=0, SIX SEED PAIRS.

Measure	Range over seed pairs
Final training CE, per seed	0.2358, 0.2374, 0.2471, 0.2489
Weight cosine (87,401 parameters)	0.023–0.071
Weight L2 distance (norms 82.8–83.8)	113.5–115.9
Same top-1 word, 8 training-text prompts	8 of 8 for every pair
Same top-1 word, 16 held-out prompts	50–69% (mean 58%)
JS of next-word dist., training-text prompts	0.052–0.060 bits
JS, held-out prompts	0.23–0.34 bits
Cosine of centred logits, held-out prompts	0.15–0.20

TABLE XXV
E2 AT FOUR SEEDS, CE-ONLY MODELS, MASK OFF: CONTRADICTED / PARSED CLAIMS.

Seed	Train prompts, T 0.8	Untrained, T 0.8	Untrained, T 1
42	7 / 598 (1.2%)	515 / 1,253 (41.1%)	522 / 1,203 (43.4%)
1	4 / 593 (0.7%)	408 / 1,151 (35.4%)	427 / 1,114 (38.3%)
2	0 / 600 (0%)	628 / 1,310 (47.9%)	601 / 1,277 (47.1%)
3	7 / 593 (1.2%)	572 / 1,243 (46.0%)	574 / 1,200 (47.8%)

defendant is the person who answers a claim. the claimant asks the court for". The verifier parsed 2 claims, both supported. A rule fired at 2 of the 20 steps (Table XXI); at the other 18 the mask changed nothing.

E. E1 and E4: four training seeds

Eight models: $\lambda_s \in \{0, 2\}$ × training seeds 42, 1, 2 and 3, all 2000 epochs, trained on all text, 68 to 81 minutes each. The measures were fixed before the last run finished. At seed 42 the λ_s=0 model has exactly the baseline's weights (L2 distance 0). *Held-out prompts:* the 16 role-and-frame prompts that never occur in the training text, the only held-out data. *Constrained positions:* the 86 training positions where a rule fires. *Four seeds:* with 4 of 4 seeds in the same direction, a paired sign-flip test cannot go below p=0.125, so "lower at every seed" is not a significance claim. Table XXII summarises

TABLE XXVI
E4 ACROSS THE SIX SEED PAIRS.

Measure	λ_s=0	λ_s=2
JS of $(P_A, 1-P_A)$, constrained (mean)	7.5e-3 (5.4e-3 to 1.0e-2)	2.5e-3 (1.9e-3 to 3.2e-3)
JS of q_A, constrained (mean)	0.0153	0.0149
Weight cosine	0.023–0.071	0.025–0.071
JS of next-word dist., held-out (mean)	0.30 bits	0.38 bits
Same top-1 word, held-out (mean)	58%	61%

the four arms.

E1 (P4): similar loss, unrelated weights, and different behaviour where training gave no data (Table XXIV). Two vectors of norm 83 at right angles are about 118 apart, so the weight vectors are close to unrelated. Part of this is symmetry (hidden units can be permuted), which is why the comparison also uses outputs. On text they trained on, all four seeds pick the same next word; on prompts outside it they pick different words a third to a half of the time. **P4 is supported.**

E2 at four seeds (P1, P3) (Table XXV). **P3 is supported:** every seed contradicts the KB on 35–48% of claims from untrained prompts. **P1 is not supported as implemented.** With the mask on, 514 of 23,664 parsed claims across all 16 mask-on cells contradict the KB. Ten cells have none, and 509 of the 514 come from seed 1 (both values of λ_s). Every one is the same case: the model writes "… is a business that **owed** money". The verifier reads a bare "owed" after a frame as *is owed*, but the mask bans "owed" only after "… that **is**", so the bare form is never masked. The mask and the verifier disagree on one surface form; the grammatical reading of "that owed money" is closer to *owes*, which the KB also bans for these roles. The fix belongs in the mask or the verifier and would change the E2 numbers, so it is left for review.

E4 (P5, P6, P7): λ_s=2 against λ_s=0 at the same seed (Tables XXIII and XXVI).

- **P5 is supported on training positions:** banned mass at the 86 constrained positions is 2.6 to 4.3 times lower at every seed, and lower on the 8 training-text prompts at every seed. On the 16 untrained prompts it is also lower at every seed, but by 0.02–0.13 out of 0.45–0.73, and contradicted claims fall by 2 to 9 points. The frequency-matched control set gets almost no mass in either arm, so the banned mass is specific to the banned verbs.
- **P6 is supported on constrained positions:** seeds disagree about three times less on how much mass goes to allowed words, while their disagreement among allowed words (q_A) is unchanged, as predicted. $H(q_A)$ falls by 2–14%, which is not collapse onto one word. Weight cosine stays near zero. On held-out prompts the seeds disagree slightly *more* with the loss (0.38 against 0.30 bits), so the reduction stays where the rules fire.
- **P7 is not testable here as stated.** On training text where no rule fires, perplexity does not rise (equal or lower at every seed). That is fit, not held-out perplexity, which needs the held-out split.
- **Overall:** at λ_s=2, the loss removes banned mass where rules fire, at no measured cost, and barely changes behaviour on contexts it never trained on. This repeats the full-length λ_s=50 run, now across four seeds.

IV. CONCLUSION

Three findings hold across seeds. **Where the constraint is trained, it holds**, at a small cost in training loss. **Beyond it, the rule does not transfer**: on unseen frames the model follows the frame, not the KB. **A guarantee covers only what its rules cover**, as the "owed" gap shows. Limits: one small model, a hand-written KB, four seeds, and no held-out text split, held-out KB or held-out cross-entropy yet.

Related work and positioning. *Constraints in the loss or output layer:* the semantic loss [6] is the term M5 adapts; semantic probabilistic layers [18] build consistency into the output layer and train through it, whereas this design keeps loss and mask separate; the pseudo-semantic loss [7] extends the idea to autoregressive models and is the precedent above. *Constrained decoding:* NeuroLogic decoding [23] searches with lexical constraints, and grammar-constrained decoding [24] restricts output to a grammar; both act only at decoding, and the M1 mask is simpler than either, a per-token ban with no lookahead. *LLMs with theorem provers:* Explanation-Refiner [25] uses a theorem prover to verify natural-language explanations and feed corrections back to an LLM; the M4 verifier is a far weaker check, a pattern parser and forward chaining over a small hand-written KB.

The same principle on any LLM. Sheth et al. [3] call systems in which an LLM hands sub-tasks to symbolic solvers *federated pipelines*. Here the KB is the solver, and the three access points of Fig. 5 become three levels of API access. With open weights, the semantic loss applies in fine-tuning. Where an API exposes logits or a logit bias, the allowed set can mask decoding. With text only, the verifier returns contradicted claims for regeneration, the verify-and-refine loop of Explanation-Refiner [25], [26] and LLM-Modulo [27]. This extension is designed, not built.

Towards neurosymbolic world models. *A proposed direction; none of this has been done, and this repository contains no robotics. Objective 1 (foundations):* move the same two-stage pipeline from token-level constraints on a toy language model to constraints on a pretrained LLM that writes action models (PDDL preconditions and effects). A planner or validator such as VAL [28], plus an entailment-based faithfulness check [25], [26], would take the verifier's role, with localised repair; GPT-4o's PDDL was 96.1% parseable but 24.8% correct [29]. O1 also tests whether world models follow plausibility over validity, like the frame effect above, and whether steering or circuit methods [30], [31] reduce it. *Objective 2 (controller):* the advert's controller "will simulate actions and verify constraints to trigger automated error recovery". That is the same split: a world model trained to respect constraints corresponds to the semantic-loss stage, and a controller that verifies constraints before acting corresponds

to the mask and verifier stage. A slow symbolic layer sets subgoals and constraints; a fast controller rolls out an action-conditioned tactile model [32], [33] and checks constraints on predicted states, so a predicted violation triggers replanning before failure, compared with post-failure recovery [34]. The main risk is the one the demo exposed: a wrong or missing rule enters training and decoding at once. So every rule stays traceable to a proof, and where formalisation fails the verifier says *unknown*.

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